

a necessary component in processors since instructions are sequenced in a pipeline. If the Branch Prediction Logic guesses the right sequence of a set of operations, then the processor will have fewer idle cycles.

Lastly, we have the concept of parallelism which is simply the way in which processors perform multiple tasks at the same time. And with this comes the concept of schedulers. Since multiple tasks can be performed by the processor, it's preferable that the processor is always active and not sitting idle. The scheduler does this by picking operations and sequencing them so that there are no (or as few) operations that have to wait. This way, if one thread in the processor has a wait time, the other thread can step in and get some work done. The concept has been in existence way before multi-core processors were introduced. That's a common misconception that we'd like to get out of the way. Let's now take a look at how the CPUs from Intel and AMD perform.

AMD'S ZEN CORE

AMD made a comeback with the Zen architecture after the disappointing Bulldozer architecture. Zen has already seen two CPU generations in the desktop space, namely the two Ryzen families.

SSH - THE ARCHITECTURE

The most prominent change in the Zen architecture is the presence of a micro-op cache. An op cache stores decoded instructions so that they can be reused again in case of repetitions. Decoding takes time and storing the decoded instructions saves time and power. Since all processors implement parallelism, you have to wait for a significant number of clock cycles for a certain instruction to be decoded before it can be executed. If there is a repetitions, then caching the decoded instruction helps saturate the instruction pipeline. Intel has had something similar for quite longer and AMD introducing a micro-op cache helped it achieve the massive IPC performance gain with Zen architecture. Without the presence of a micro-op cache, the instructions can still be cache but

CACHE LEVELS		
Architecture	Zen	Skylake
L1-I	64KB 4-way	32KB 8-way
L1-D	32KB 8-way Write Back	32KB 8-way Write Back
L2	512KB 8-way	256KB 4-way
L3	1-2MB/Core 16-way	2MB/Core 16-way

it'll be from the colder caches and that takes more time to refetch as well.

Companies don't reveal all the changes made to processors, in this case, the branch prediction logic was improved but we have no idea exactly what was improved. Then there's the instruction decoder which is capable of decoding 4 instructions per clock cycle. So the scheduler should have access to 4 decoded operations per clock, right? Nope. Remember the micro-op cache we spoke of? Throw that in the mix and you can have 6 decoded operations per clock cycle.

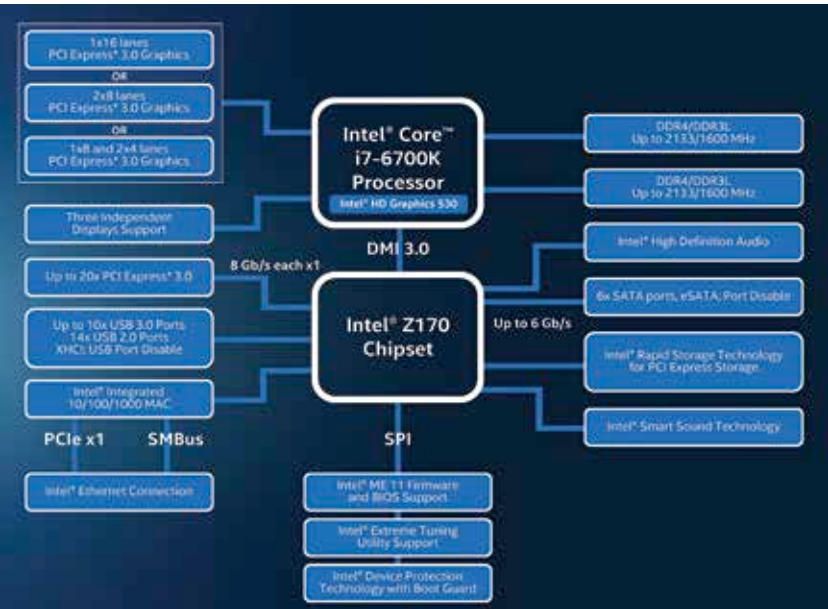
AMD has separate schedulers for the Integer and the Floating Point data. In this particular block diagram, we can see six integer schedulers whereas there's only one floating point scheduler. This is because there are a lot more integer values that need to be processed compared to floating point values. So having more dedicated resources for integers helps execute more instructions per clock (IPC). And

IPC is a proper measure of a processors performance.

The ALU as well as the AGU (Address Generation Unit) (load and store) operations are handled on the INT side of the core while the FP side has the two ADD (Addition) and MUL (Multiply) ports. The reason for having these on the FP side is to facilitate a 256-bit AVX operation per clock cycle. AVX is a powerful Instruction Set used in modern processors. The latest addition to this would be the AVX-512 which is at the moment only present on Intel processors.

SSH - THE CACHE HIERARCHY

One of the other major changes to the Zen architecture was the change in cache hierarchy. The L1 cache has now doubled in size and the associativity is also greater. Cache memory gets smaller as you get closer to the cores. So there's always the issue of multiple cores vying for the same memory space in the case of a shared cache and



The Intel Skylake block diagram